PRACTICAL COURSE WORKBOOK

SQL & Relational Databases

Model, query and explain dependable data

LEVEL	Beginner
GUIDED TIME	5 hours across 12 lessons
PROJECT	a working data model with validated queries
SKILLS	SQL, PostgreSQL, MySQL, Joins, Indexing, Critical evaluation
EDITION	Structured draft - version 0.9

WHAT YOU WILL BE ABLE TO DO

- Use SQL appropriately in a realistic, bounded task.
- Use PostgreSQL appropriately in a realistic, bounded task.
- Use MySQL appropriately in a realistic, bounded task.
- Use Joins appropriately in a realistic, bounded task.

WHO THIS IS FOR

Developers and analysts who need dependable data systems.

BEFORE YOU BEGIN

- Basic programming or spreadsheet experience

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

A clear route through the course

SQL & Relational Databases is a structured, practical course covering SQL, PostgreSQL, MySQL, Joins, Indexing. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 SQL: Data modelling	1. Entities with SQL 2. Relationships with PostgreSQL 3. Constraints with MySQL
02 PostgreSQL: Querying	1. Selection with Joins 2. Joins with Indexing 3. Aggregation with SQL
03 MySQL: Reliability	1. Transactions with PostgreSQL 2. Indexes with MySQL 3. Backups with Joins
04 Joins: Production design	1. Security with Indexing 2. Performance with SQL 3. Operations with PostgreSQL

HOW TO USE EACH LESSON PAGE

1. Read the objective and identify the decision it supports.
2. Attempt the retrieval prompt before checking the online explanation.
3. Reproduce the worked example with the stated input.
4. Test one normal case and one boundary or failure case.
5. Record the result, limitation and next action in the evidence log.

PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

SQL: Data modelling

Apply SQL through entities, relationships, constraints.

LESSON 01 / 18 MIN

Entities with SQL

Learning objectives

- Explain entities in the context of SQL & Relational Databases.
- Apply SQL to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

entities, SQL, databases

Retrieval prompt

In SQL & Relational Databases, which evidence best supports a entities result produced with SQL?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 02 / 22 MIN

Relationships with PostgreSQL

Learning objectives

- Explain relationships in the context of SQL & Relational Databases.
- Apply PostgreSQL to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

relationships, PostgreSQL, databases

Retrieval prompt

In SQL & Relational Databases, which evidence best supports a relationships result produced with PostgreSQL?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 03 / 26 MIN

Constraints with MySQL

Learning objectives

- Explain constraints in the context of SQL & Relational Databases.
- Apply MySQL to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

constraints, MySQL, databases

Retrieval prompt

In SQL & Relational Databases, which evidence best supports a constraints result produced with MySQL?

Evidence to keep	[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

PostgreSQL: Querying

Apply PostgreSQL through selection, joins, aggregation.

LESSON 04 / 30 MIN

Selection with Joins

Learning objectives

- Explain selection in the context of SQL & Relational Databases.
- Apply Joins to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

selection, Joins, databases

Retrieval prompt

In SQL & Relational Databases, which evidence best supports a selection result produced with Joins?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 05 / 18 MIN

Joins with Indexing

Learning objectives

- Explain joins in the context of SQL & Relational Databases.
- Apply Indexing to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

joins, Indexing, databases

Retrieval prompt

In SQL & Relational Databases, which evidence best supports a joins result produced with Indexing?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 06 / 22 MIN

Aggregation with SQL

Learning objectives

- Explain aggregation in the context of SQL & Relational Databases.
- Apply SQL to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

aggregation, SQL, databases

Retrieval prompt

In SQL & Relational Databases, which evidence best supports a aggregation result produced with SQL?

Evidence to keep	[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE
Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY
Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW
Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 03

MySQL: Reliability

Apply MySQL through transactions, indexes, backups.

LESSON 07 / 26 MIN

Transactions with PostgreSQL

Learning objectives

- Explain transactions in the context of SQL & Relational Databases.
- Apply PostgreSQL to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

transactions, PostgreSQL, databases

Retrieval prompt

In SQL & Relational Databases, which evidence best supports a transactions result produced with PostgreSQL?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 08 / 30 MIN

Indexes with MySQL

Learning objectives

- Explain indexes in the context of SQL & Relational Databases.
- Apply MySQL to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

indexes, MySQL, databases

Retrieval prompt

In SQL & Relational Databases, which evidence best supports a indexes result produced with MySQL?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 09 / 18 MIN

Backups with Joins

Learning objectives

- Explain backups in the context of SQL & Relational Databases.
- Apply Joins to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

backups, Joins, databases

Retrieval prompt

In SQL & Relational Databases, which evidence best supports a backups result produced with Joins?

Evidence to keep	[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 04

Joins: Production design

Apply Joins through security, performance, operations.

LESSON 10 / 22 MIN

Security with Indexing

Learning objectives

- Explain security in the context of SQL & Relational Databases.
- Apply Indexing to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

security, Indexing, databases

Retrieval prompt

In SQL & Relational Databases, which evidence best supports a security result produced with Indexing?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 11 / 26 MIN

Performance with SQL

Learning objectives

- Explain performance in the context of SQL & Relational Databases.
- Apply SQL to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

performance, SQL, databases

Retrieval prompt

In SQL & Relational Databases, which evidence best supports a performance result produced with SQL?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 12 / 30 MIN

Operations with PostgreSQL

Learning objectives

- Explain operations in the context of SQL & Relational Databases.
- Apply PostgreSQL to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

operations, PostgreSQL, databases

Retrieval prompt

In SQL & Relational Databases, which evidence best supports a operations result produced with PostgreSQL?

Evidence to keep	[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

CAPSTONE PROJECT

Produce a reviewable SQL & Relational Databases project using SQL, PostgreSQL, MySQL.

DELIVERABLE	A working SQL & Relational Databases artefact plus an evidence-based self-review.
TOOLS	A suitable SQL environment, A plain-text decision log, Test data or realistic sample material

Production plan

1. Define the intended user, outcome and constraints.
2. Create the smallest complete result using SQL.
3. Test one normal case, one boundary case and one failure response.
4. Revise the work from the evidence and preserve before-and-after results.
5. Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

PROJECT EVIDENCE

Document decisions another person can review

1. PURPOSE AND USER
State the intended user, need, expected outcome and constraints.

2. INPUTS AND ASSUMPTIONS
List source files, data, versions, permissions and assumptions.

PROJECT EVIDENCE

Tests, limitations and revision

3. NORMAL CASE

Expected result, observed result and evidence location.

4. BOUNDARY OR FAILURE CASE

What was difficult, what happened and why it matters.

5. REVISION AND LIMITATION

What changed, what improved and what remains unproven.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

FINAL DECISION
Is the project ready to share? State the evidence, remaining risk, owner and next review date.

Authoritative references

PostgreSQL Tutorial

PostgreSQL Global Development Group - reviewed 2026-08-21

https://www.postgresql.org/docs/current/tutorial.html

SQL Language

PostgreSQL Global Development Group - reviewed 2026-08-21

https://www.postgresql.org/docs/current/sql.html

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the SQL scope.